

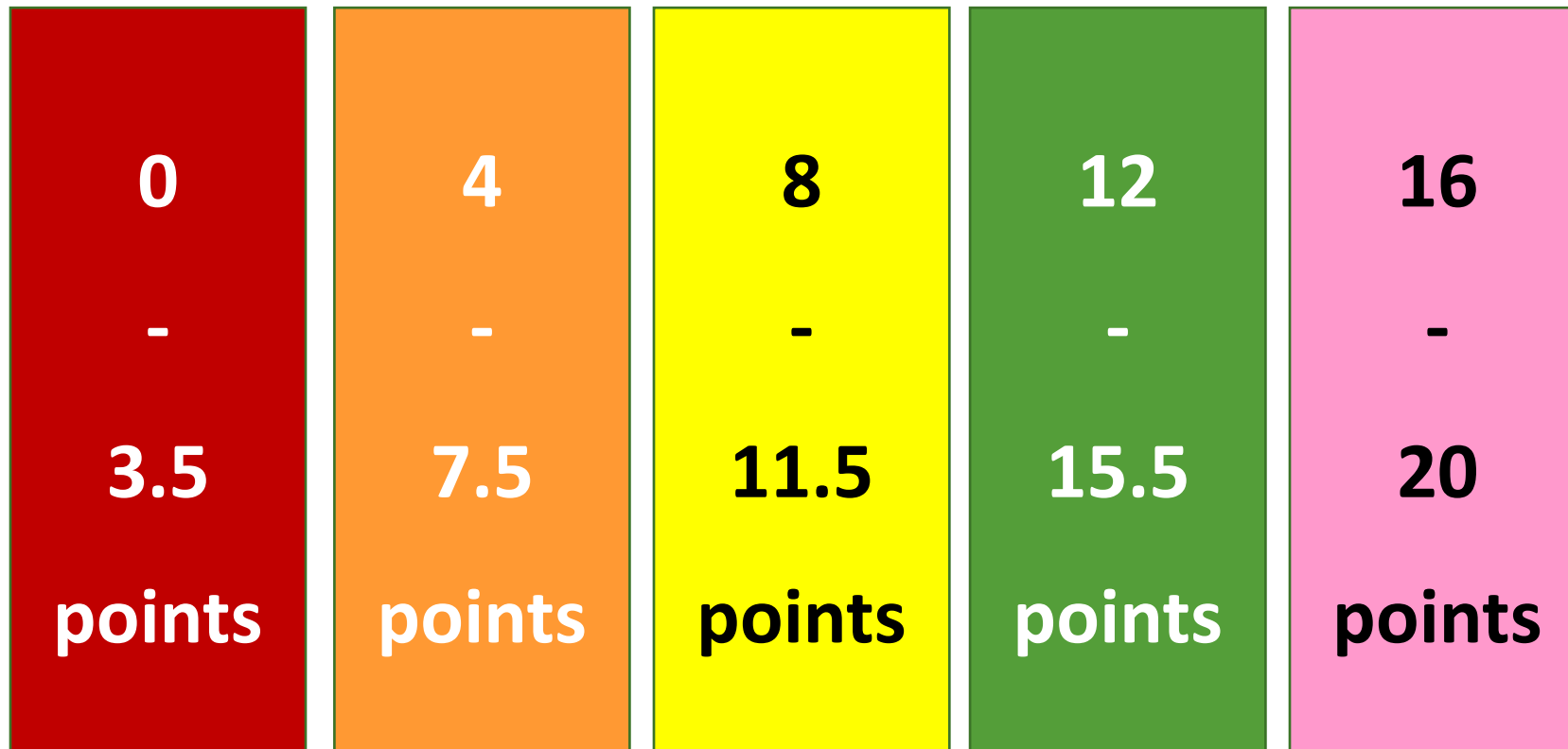
Lec13A

Revision 02 - Linear Maps

The Wooclap scores are available on NTULearn.

Which box do you belong to?

To obtain one point, you need to answer TRUTHFULLY.



QUESTION 1.

(20 marks)

Let $T : \mathcal{M}_{3 \times 3}(\mathbb{R}) \rightarrow \mathcal{M}_{3 \times 3}(\mathbb{R})$ be the function defined by $T(M) = M - M^T$.

(a) Show that T is a linear operator. $\rightarrow$ linear map

(b) Consider the following nine matrices in $\mathcal{M}_{3 \times 3}(\mathbb{R})$:

$$\begin{aligned} A &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & B &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & C &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \\ D &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & E &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, & F &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \\ G &= \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & H &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, & N &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}. \end{aligned}$$

Let $\mathcal{B} = \{A, B, C, D, E, F, G, H, N\}$. Show that $\mathcal{B}$ is a basis for $\mathcal{M}_{3 \times 3}(\mathbb{R})$.

(c) Determine the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$.

(d) Determine a basis for $\text{null}(T)$.

(e) Determine a basis for $\text{range}(T)$.

Let $T : \mathcal{M}_{3 \times 3}(\mathbb{R}) \rightarrow \mathcal{M}_{3 \times 3}(\mathbb{R})$ be the function defined by $T(M) = M - M^T$.

(a) Show that T is a linear operator.

What are the properties that you need to verify?

- ~~1.~~ T is additive
- ~~2.~~ T is homogeneous
3. $\text{null}(T) = \{\mathbf{0}\}$
4. $\text{range}(T) = \text{codomain}$
5. $\text{domain} = \text{codomain}$

Let $T : \mathcal{M}_{3 \times 3}(\mathbb{R}) \rightarrow \mathcal{M}_{3 \times 3}(\mathbb{R})$ be the function defined by $T(M) = M - M^T$.

(a) Show that T is a linear operator.

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To show that T is additive, what must you show?

1. $T(M + N) = T(M) + T(N)$ for all M, N . * For all $\hat{M}, \hat{N} \in \mathcal{M}_{3 \times 3}$.
2. $(S + T)(M) = S(M) + T(M)$ for all S, T .

Proof

$$\begin{aligned}\uparrow(\hat{M} + \hat{N}) &= (\hat{M} + \hat{N}) - (\hat{M} + \hat{N})^T \\ &= (\hat{M} + \hat{N}) - (\hat{M}^T + \hat{N}^T) \\ &= (\hat{M} - \hat{M}^T) + (\hat{N} - \hat{N}^T) \\ &= \uparrow(\hat{M}) + \uparrow(\hat{N})\end{aligned}$$

$\therefore \uparrow$ is additive

Let $T : \mathcal{M}_{3 \times 3}(\mathbb{R}) \rightarrow \mathcal{M}_{3 \times 3}(\mathbb{R})$ be the function defined by $T(M) = M - M^T$.

(a) Show that T is a linear operator.

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Can you show that T is homogeneous?

(b) Consider the following nine matrices in $\mathcal{M}_{3 \times 3}(\mathbb{R})$:

$$\begin{aligned} A &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & B &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & C &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \\ D &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & E &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, & F &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \\ G &= \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & H &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, & N &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}. \end{aligned}$$

Let $\mathcal{B} = \{A, B, C, D, E, F, G, H, N\}$. Show that $\mathcal{B}$ is a basis for $\mathcal{M}_{3 \times 3}(\mathbb{R})$.

To show that $\mathcal{B}$ is a basis, what must you show?

1. $\mathcal{B}$ is linearly independent and $\text{span}(\mathcal{B}) = \mathcal{M}_{3 \times 3}(\mathbb{R})$.
 2. $\mathcal{B}$ is linearly independent and $|\mathcal{B}| = 9$.
 3. $\text{span}(\mathcal{B}) = \mathcal{M}_{3 \times 3}(\mathbb{R})$ and $|\mathcal{B}| = 9$.
- } all 3 are correct

(b) Consider the following nine matrices in $\mathcal{M}_{3 \times 3}(\mathbb{R})$:

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & \mathbf{B} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & \mathbf{C} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \\ \mathbf{D} &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & \mathbf{E} &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, & \mathbf{F} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \\ \mathbf{G} &= \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & \mathbf{H} &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, & \mathbf{N} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}. \end{aligned}$$

Let $\mathcal{B} = \{\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}, \mathbf{E}, \mathbf{F}, \mathbf{G}, \mathbf{H}, \mathbf{N}\}$. Show that $\mathcal{B}$ is a basis for $\mathcal{M}_{3 \times 3}(\mathbb{R})$.

Consider

$$\lambda_A \mathbf{A} + \lambda_B \mathbf{B} + \cdots + \lambda_N \mathbf{N} = \mathbf{0} - (\star)$$

To show that $\mathcal{B}$ is linearly independent, what must you show?

1. $(\star)$ implies that $\lambda_A = \lambda_B = \cdots = \lambda_N = 0$.
2. $\lambda_A = \lambda_B = \cdots = \lambda_N = 0$ implies $(\star)$.

(b) Consider the following nine matrices in $\mathcal{M}_{3 \times 3}(\mathbb{R})$:

$$\begin{aligned} A &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & B &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & C &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \\ D &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & E &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, & F &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \\ G &= \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, & H &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, & N &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}. \end{aligned}$$

Let $\mathcal{B} = \{A, B, C, D, E, F, G, H, N\}$. Show that $\mathcal{B}$ is a basis for $\mathcal{M}_{3 \times 3}(\mathbb{R})$.

$$\lambda_A \mathbf{A} + \lambda_B \mathbf{B} + \dots + \lambda_N \mathbf{N} = \begin{pmatrix} \lambda_A & \lambda_D + \lambda_G & \lambda_E \\ \lambda_D - \lambda_G & \lambda_B & \lambda_F \\ \lambda_E - \lambda_H & \lambda_F - \lambda_N & \lambda_C \end{pmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\lambda_A = \lambda_B = \lambda_C = 0$$

$$\begin{aligned} \lambda_D - \lambda_G &= 0 \\ \lambda_D + \lambda_G &= 0 \end{aligned} \Rightarrow \begin{cases} \lambda_D = \lambda_G = 0 \end{cases}$$

$$\text{Similarly } \lambda_E = \lambda_H = \lambda_N = \lambda \neq 0$$

$$\therefore (\hat{A} \dots \hat{N}) \text{ is linearly independent}$$

Let $T : \mathcal{M}_{3 \times 3}(\mathbb{R}) \rightarrow \mathcal{M}_{3 \times 3}(\mathbb{R})$ be the function defined by $T(M) = M - M^T$.

(c) Determine the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$.

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Can you determine $[T]_{\mathcal{B}}^{\mathcal{B}}$?

$$T(\hat{A}) = 0$$

$$T(\hat{B}) = 0$$

$$T(\hat{C}) = 0$$

$$T(\hat{D}) = 0$$

$$T(\hat{E}) = 0$$

$$T(\hat{F}) = 0$$

$$T(\hat{G}) = \begin{bmatrix} 2 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 2\hat{G}$$

$$T(\hat{H}) = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} = 2\hat{H}$$

$$T(\hat{N}) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 2\hat{N}$$

$$[T]_{\mathcal{B}}^{\mathcal{B}} =$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

no. of rows = no. of elements in basis of codomain = $\dim(\text{codomain})$
 # cols = # elements in basis of domain = $\dim(\text{domain})$
 represents codomain
 represents domain

Let $T : \mathcal{M}_{3 \times 3}(\mathbb{R}) \rightarrow \mathcal{M}_{3 \times 3}(\mathbb{R})$ be the function defined by $T(M) = M - M^T$.

- (c) Determine the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$.
- (d) Determine a basis for $\text{null}(T)$.
- (e) Determine a basis for $\text{range}(T)$.

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What is the reduced row echelon form of $[T]_{\mathcal{B}}^{\mathcal{B}}$?

✓ 1.
$$\begin{bmatrix} 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 1 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 1 \end{bmatrix}$$

2.
$$\begin{bmatrix} 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \end{bmatrix}$$

Let $T : \mathcal{M}_{3 \times 3}(\mathbb{R}) \rightarrow \mathcal{M}_{3 \times 3}(\mathbb{R})$ be the function defined by $T(M) = M - M^T$.

(d) Determine a basis for $\text{null}(T)$.

$$\text{null}([T]_B^B) = \{s_1, s_2, \dots, s_6, 0, 0, 0\} : s_i \in \mathbb{R}\}$$

$$\text{Basis for } \text{null}(T) = [\hat{A}, \hat{B}, \hat{C}, \hat{D}, \hat{E}, \hat{F}]$$

$$\begin{bmatrix} 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 1 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 1 \end{bmatrix}$$

$s_1 \quad s_2 \quad \dots \quad s_6 \quad \triangle \quad \triangle \quad \triangle$

Let $T : \mathcal{M}_{3 \times 3}(\mathbb{R}) \rightarrow \mathcal{M}_{3 \times 3}(\mathbb{R})$ be the function defined by $T(M) = M - M^T$.

(e) Determine a basis for $\text{range}(T)$.

$$\begin{aligned} \text{Basis for range}(T) &= [T(\hat{G}), T(\hat{H}), T(\hat{N})] \\ &= [2\hat{G}, 2\hat{H}, 2\hat{N}] \end{aligned}$$

$$\begin{bmatrix} 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 1 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 1 \end{bmatrix}$$

pivots

QUESTION 5.

(20 marks)

Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces.

"Proof Question"

- (a) Show that there exists a surjective linear map from $\mathcal{V}$ to $\mathcal{W}$ if and only if $\dim(\mathcal{V}) \geq \dim(\mathcal{W})$.
- (b) Let $\mathcal{U}$ be a subspace of $\mathcal{V}$. Show that there exists a linear map T from $\mathcal{V}$ to $\mathcal{W}$ with $\text{null}(T) = \mathcal{U}$ if and only if $\dim(\mathcal{U}) \geq \dim(\mathcal{V}) - \dim(\mathcal{W})$.

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map with $\dim(\mathcal{V}) = n$ and $\dim(\mathcal{W}) = m$.

What do we know?

1. $\dim \text{null}(T) + \dim \text{range}(T) = n$
2. $\dim \text{null}(T) + \dim \text{range}(T) = m$

Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces.

$$\dim(\mathcal{W}) = m$$

- (a) Show that there exists a surjective linear map from $\mathcal{V}$ to $\mathcal{W}$ if and only if $\dim(\mathcal{V}) \geq \dim(\mathcal{W})$.

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Argument 1. Let T be a surjective map.

Then $\dim \text{range}(T) = m$ and so, $\dim \text{null}(T) + m = n$.

Hence, $n \geq m$ or $\dim(\mathcal{V}) \geq \dim(\mathcal{W})$.

What does Argument 1 prove?

1. There exists a surjective map from $\mathcal{V}$ to $\mathcal{W}$ implies $\dim(\mathcal{V}) \geq \dim(\mathcal{W})$.
2. $\dim(\mathcal{V}) \geq \dim(\mathcal{W})$ implies there exists a surjective map from $\mathcal{V}$ to $\mathcal{W}$.

Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces.

- (a) Show that there exists a surjective linear map from $\mathcal{V}$ to $\mathcal{W}$ if and only if $\dim(\mathcal{V}) \geq \dim(\mathcal{W})$.

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Argument 2. Suppose $\dim(\mathcal{V}) < \dim(\mathcal{W})$ and T is a surjective map.

Proof by contradiction

Then $\dim \text{range}(T) = m$ and so, $\dim \text{null}(T) + m = n$.

Hence, $n \geq m$ contradicting the fact $\dim(\mathcal{V}) < \dim(\mathcal{W})$.

Therefore, T is not a surjective map.

What does Argument 2 prove?

1. There exists a surjective map from $\mathcal{V}$ to $\mathcal{W}$ implies $\dim(\mathcal{V}) \geq \dim(\mathcal{W})$.
- ~~2.~~ $\dim(\mathcal{V}) \geq \dim(\mathcal{W})$ implies there exists a surjective map from $\mathcal{V}$ to $\mathcal{W}$.

Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces.

- (a) Show that there exists a surjective linear map from $\mathcal{V}$ to $\mathcal{W}$ if and only if $\dim(\mathcal{V}) \geq \dim(\mathcal{W})$.

Assume $\dim(\mathcal{V}) \geq \dim(\mathcal{W})$.

We construct a surjective linear map $\hat{T}: \mathcal{V} \rightarrow \mathcal{W}$

Let $[x_1, \dots, x_m, \dots, x_n]$ be a basis for $\mathcal{V}$.

Let $[w_1, \dots, w_m]$ be a basis for $\mathcal{W}$.

Then there is a linear map $\hat{T}$ s.t. $\hat{T}(x_1) = w_1, \hat{T}(x_2) = w_2, \dots, \hat{T}(x_m) = w_m$
 $\hat{T}(x_{m+1}) = \hat{T}(x_{m+2}) = \dots = \hat{T}(x_n) = 0.$

It remains to check that $\hat{T}$ is surjective.

$$\begin{aligned} \text{range}(\hat{T}) &= \text{span}[\hat{T}(x_1), \dots, \hat{T}(x_m), \dots, \hat{T}(x_n)] \\ &= \text{span}(w_1, \dots, w_m) = \mathcal{W}. \end{aligned}$$

Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces.

- (b) Let $\mathcal{U}$ be a subspace of $\mathcal{V}$. Show that there exists a linear map $\mathbf{T}$ from $\mathcal{V}$ to $\mathcal{W}$ with $\text{null}(\mathbf{T}) = \mathcal{U}$ if and only if $\dim(\mathcal{U}) \geq \dim(\mathcal{V}) - \dim(\mathcal{W})$.

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